

Prepare R&D Document on How to setup Site to Site using Hyper-V

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1. Introduction

A Site-to-Site VPN (Virtual Private Network) allows two geographically separated networks (usually offices or data centers) to securely communicate over the internet. This is critical for organizations that need to securely connect different branch offices, backup sites, or data centers.

Hyper-V, a native hypervisor by Microsoft, allows you to create and manage virtual machines (VMs). By using Windows Server in Hyper-V and configuring RRAS, we can simulate and test a Site-to-Site VPN setup between two virtual networks.

2. Objective

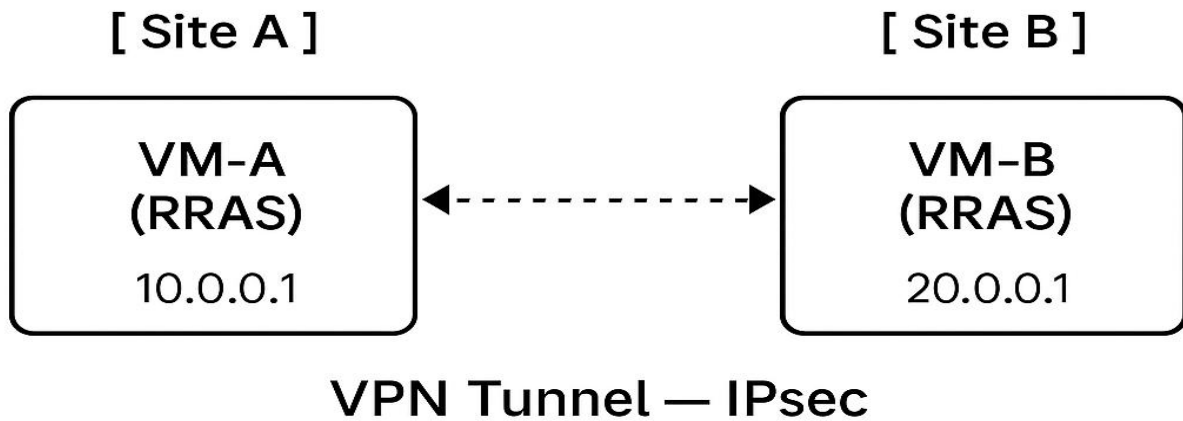
The objective of this assignment is to successfully configure and test a Site-to-Site VPN between two virtual networks (VNet-A and VNet-B) using Windows Server VMs in Hyper-V, ensuring:

- Encrypted communication between the two sites
- Successful ping/access to remote network resources
- Proper routing and tunnel establishment using RRAS

3. Prerequisites

Component	Requirement
Host Machine	Windows 10/11 Pro or Enterprise (Hyper-V enabled)
Hyper-V Virtual Machines	2x Windows Server 2016/2019 VMs
Roles & Features	RRAS (Routing and Remote Access Service)
Static IP Configuration	Required on both VMs
Internet Connectivity	Required for testing VPN tunneling
Administrator Privileges	Required on all VMs and host system

4. Diagram



5. Step-by-Step Procedure

STEP 1: Enable Hyper-V on Host Machine

1. Go to “Turn Windows Features on or off”
2. Enable Hyper-V, Hyper-V Management Tools, and Hyper-V Platform
3. Restart your system

STEP 2: Create Two Virtual Machines (VMs)

1. Open Hyper-V Manager
2. Create:
 - VM-SiteA with IP 10.0.0.1
 - VM-SiteB with IP 20.0.0.1
3. Install Windows Server 2016/2019 on both

STEP 3: Set Static IP Addresses

Inside each VM:

- Site A:
 - IP: 10.0.0.1

- Subnet: 255.255.255.0
- Gateway: 10.0.0.254 (optional)
- Site B:
 - IP: 20.0.0.1
 - Subnet: 255.255.255.0
 - Gateway: 20.0.0.254 (optional)

STEP 4: Install RRAS on Both VMs

1. Open Server Manager > Add Roles and Features
2. Select Remote Access
3. Under Role Services, select Routing and DirectAccess and VPN (RAS)
4. Complete installation and restart VM

STEP 5: Configure RRAS for VPN on Site A

1. Open Routing and Remote Access
2. Right-click server > "Configure and Enable Routing and Remote Access"
3. Select Custom Configuration
4. Choose VPN access and LAN Routing
5. Start RRAS service

STEP 6: Add Static Routes (Optional)

In RRAS:

- Go to IPv4 > Static Routes
- Add route to the remote network
 - For Site A: Route to 20.0.0.0/24 via VPN interface
 - For Site B: Route to 10.0.0.0/24 via VPN interface

STEP 7: Create Demand-Dial Interface (VPN Tunnel)

On both VMs:

1. In RRAS > Right-click Network Interfaces > New Demand-Dial Interface
2. Interface name: SiteBLink (on SiteA), SiteALink (on SiteB)
3. Choose Connect using VPN
4. VPN Type: IKEv2 or L2TP/IPSec
5. Enter IP address of remote site (public or virtual IP of VM)

6. Choose Route IP Packets on this interface
7. Enter static route for remote network (e.g., 20.0.0.0/24)
8. Set up credentials (must match at both ends)

STEP 8: Configure VPN Policies and Firewall

- Ensure firewall rules allow VPN protocols (IKE, L2TP, IPsec)
- Open UDP 500, UDP 4500, and ESP (IP Protocol 50)
- Configure Windows Firewall to allow RRAS and VPN traffic

STEP 9: Test the VPN Connection

1. Start Demand-Dial Interface from Site A
2. Check RRAS logs and status
3. From Site A, try pinging Site B (e.g., ping 20.0.0.1)
4. Do the reverse test from Site B to Site A

8. Benefits of this Setup

- Cost-Effective: No need for physical routers or leased lines
- Safe Testing: Hyper-V allows VPN testing in a virtual environment
- Skill Building: Develops hands-on experience with RRAS and Windows Server
- Scalability: Can add more sites using similar steps

9. Limitations

- Works only for lab/testing unless VMs are exposed to real public IPs
- RRAS lacks some advanced enterprise VPN features
- Needs careful configuration to avoid IP conflicts

10. Conclusion

This assignment successfully demonstrated how to configure a Site-to-Site VPN using Hyper-V with RRAS on Windows Server. The process covered creating virtual networks, assigning IPs, configuring demand-dial interfaces, and ensuring connectivity through a secure tunnel. This simulation is valuable for learning VPN architecture in a safe and isolated test environment.